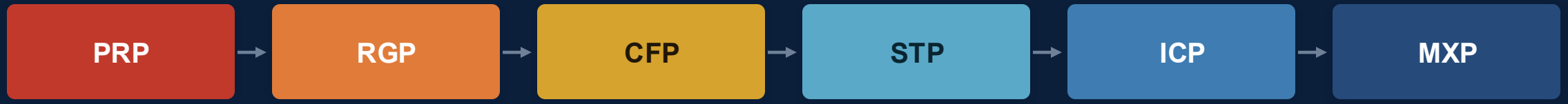
CRYOGENIC THERMAL MODELLING

Pulse-Tube Cooldown Simulator

A lumped-capacitance nodal thermal model for dilution-refrigerator precooling — and the systematic physics added to it.

Python · NiceGUI · adaptive-timestep transient solver

6-STAGE THERMAL CHAIN (WARM → COLD)



What the model already did well

The baseline was already a credible, physically-grounded solver — the additions build on solid ground.

Coupled nodal solver

Energy balance on every stage with temperature-dependent heat capacity.

Cryo material database

Cp(T) and k(T) for stainless, OFHC copper, G-10 and aluminium alloys.

Pulse-tube capacity curves

Measured 1st- and 2nd-stage heat-lift Q(T) drive the cooling.

Gas-gap conduction

Viscous ⊕ free-molecular harmonic blend across the whole transition.

Radiative shield cascades

Stefan–Boltzmann with series emissivity for nested shields.

Parasitics + MLI

Support-strut conduction and multilayer-insulation radiation loads.

Linked auxiliary masses

Discrete masses — heat exchangers, brackets — as their own nodes.

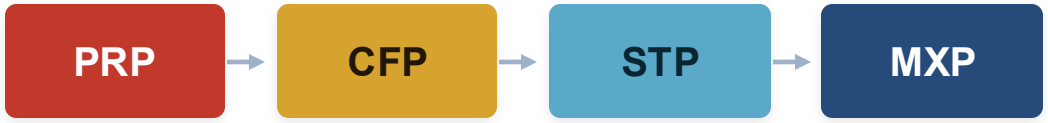
Adaptive solver + GUI

Stable transients, a NiceGUI front-end, and saved system profiles.

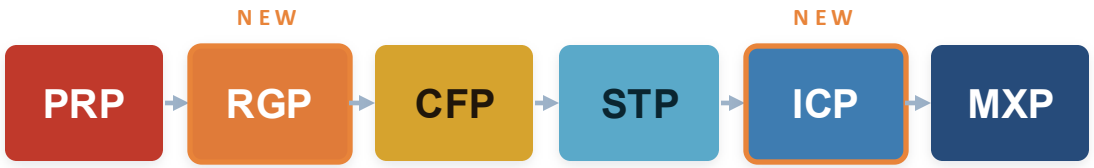
IMPROVEMENT 01

From four stages to six

BEFORE — 4 stages



AFTER — 6 stages



RGP (regenerator plate) and **ICP** (inter-stage plate) added as new intercept nodes.

Two intercept stages turn the chain into six coupled nodes. Old four-stage profiles still load: the single PRP–CFP gap is aliased into PRP–RGP + RGP–CFP (and STP–MXP likewise), so nothing breaks on open.

GOVERNING BALANCE — PER NODE i

$$C_i(T_i)\frac{dT_i}{dt} = -\dot{Q}_{\mathrm{cool},i} + \sum_j \dot{Q}_{ij}^{\mathrm{gas}} + \dot{Q}_i^{\mathrm{rad}} + \dot{Q}_i^{\mathrm{cond}} + \dot{Q}_i^{\mathrm{lm}}$$

with temperature-dependent heat capacity

$$C_i(T) = \sum_k m_k\, c_{p,k}(T)$$

IMPROVEMENT 02

Regenerator-plate cooling & the missing matrix mass

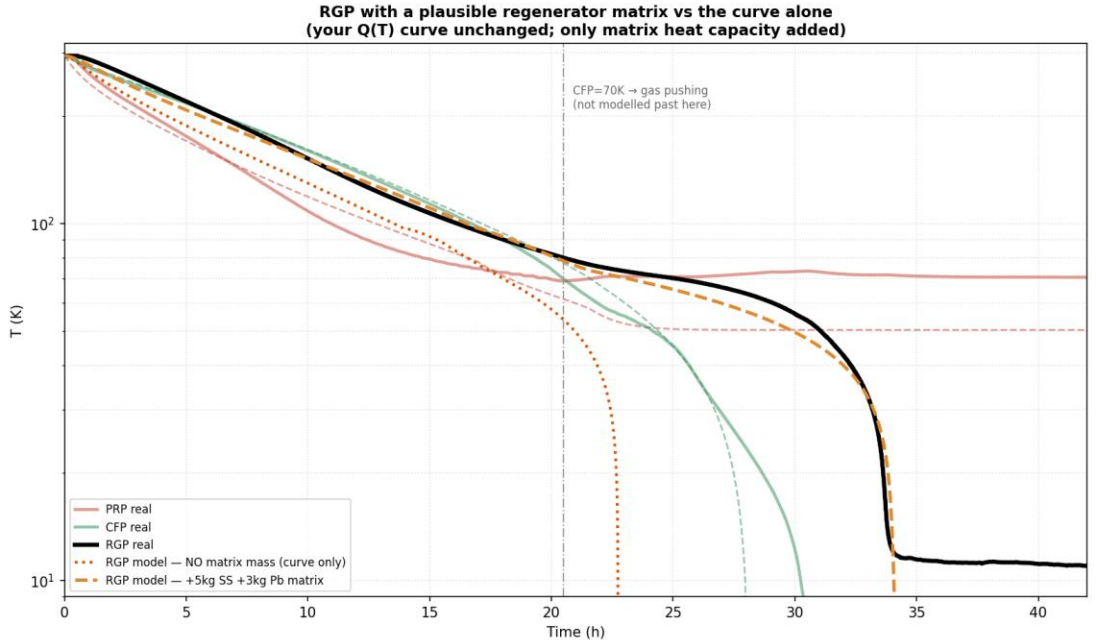
A dedicated regenerator intercept was added — its own heat-lift curve and a no-load floor. **But the decisive physics was regenerator matrix heat capacity.** RGP is coupling-dominated, so the curve barely sets its descent; adding the matrix as a thermal-mass node makes the plate lag exactly as measured.

INTERCEPT HEAT-LIFT

$$\dot{Q}_{\text{reg}} = f_R(T) \text{ from } (T_R,\, Q_R), \quad T \geq T_{\text{base, reg}}$$

ADDED MATRIX NODE — THE MISSING TERM

$$C_{\text{matrix}}(T) = \sum_k m_k\, c_{p,\,k}(T)$$



Same Q(T) curve — only matrix heat capacity added. RGP tracks the real trace down onto its 10.6 K plateau.

IMPROVEMENT 03

Physically-grounded gas-gap heat switches

The hard binary shutoff was replaced by adsorption physics. A sorption pump sets the gap pressure through a Dubinin–Radushkevich isotherm, and the switch conductance follows from that pressure via the viscous / free-molecular blend.

DR predicts shutoff within ~10 K of measurement — independent corroboration, not a fit. Precomputed K(T) tables let a full external switch model drop straight in.

T_sor → DR isotherm → gap pressure → conductance blend → K_eff

GAS CONDUCTION — BOTH REGIMES, BLENDED

$$\dot{Q}_{\mathrm{fm}} = \alpha A P \sqrt{\frac{R}{2\pi M \bar{T}}}\,\frac{\gamma+1}{2(\gamma-1)}\,(T_h - T_c)$$

$$\dot{Q}_{\mathrm{vis}} = \frac{k_g(\bar{T})\,A}{L}\,(T_h - T_c)$$

$$\dot{Q}_{\mathrm{gas}} = \frac{\dot{Q}_{\mathrm{fm}}\,\dot{Q}_{\mathrm{vis}}}{\dot{Q}_{\mathrm{fm}} + \dot{Q}_{\mathrm{vis}}} = \left(\dot{Q}_{\mathrm{fm}}^{-1} + \dot{Q}_{\mathrm{vis}}^{-1}\right)^{-1}$$

ADSORPTION SETS THE PRESSURE (DUBININ–RADUSHKEVICH)

$$\theta = \frac{q}{q_{\max}} = \exp\left[-\left(\frac{\varepsilon}{E_0}\right)^2\right]$$

$$\varepsilon = RT\ln\left(\frac{P_0}{P}\right) \qquad E_0 = E_{0a} + E_{0b}\,T$$

IMPROVEMENT 04

Exchange-gas precooling

A small gas charge left in the vacuum can during cooldown couples the stages by gas conduction, then is pumped out once CFP reaches the cutoff. Its pressure follows an empirical cubic in CFP temperature; the same viscous / free-molecular blend does the conduction — and it now reaches the linked masses, not just the plates.

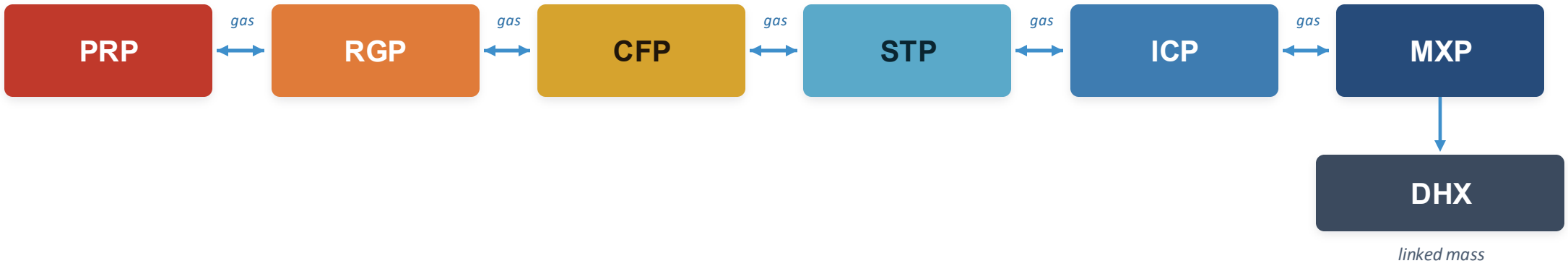
OVC PRESSURE vs CFP TEMPERATURE

$$P_{xg}(T_{CFP}) = c_3 T^3 + c_2 T^2 + c_1 T + c_0 \quad \text{(torr)}$$

CONDUCTION AT THAT PRESSURE

$$\dot{Q}_{xg} = \frac{\dot{Q}_{fm}\,\dot{Q}_{vis}}{\dot{Q}_{fm} + \dot{Q}_{vis}} \quad \text{at} \quad P = P_{xg}$$

OVC GAS COUPLES NEIGHBOURS + LINKED MASSES



Pumped to full vacuum once CFP ≤ 70 K — after that the model is pure conduction and radiation again.

IMPROVEMENT 05

Bolted-joint contact resistance

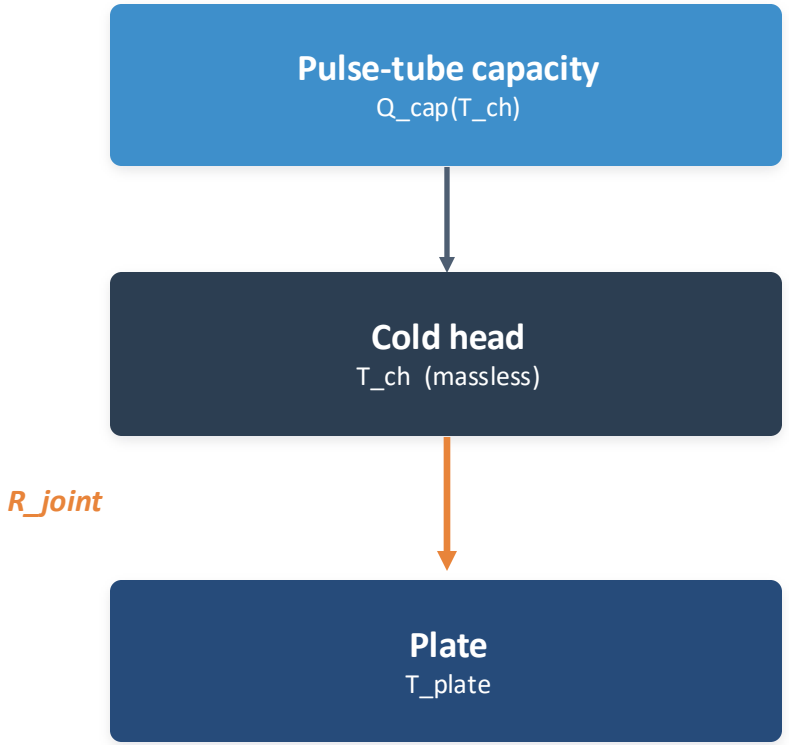
The capacity curve is no longer applied directly at the plate. **A massless cold-head node is joined to the plate through a bracket contact resistance** and settles where the capacity balances the conducted heat. A single `contact_factor` scales the joint — at zero it collapses exactly onto the original direct-capacity behaviour.

COLD-HEAD SETTLES AT THE BALANCE POINT

$$\dot{Q}_{\text{cap}}(T_{\text{ch}}) = \frac{T_{\text{plate}} - T_{\text{ch}}}{R_{\text{joint}}}$$

JOINT RESISTANCE (SCALED BY contact_factor)

$$R_{\text{joint}} = f_c \, r_c(T) \sum_i \frac{n_i}{A_i}$$



VALIDATION

Calibrated against real hardware

